

Fairness and Unemployment

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There have been natural reasons for the recent development and partial acceptance of efficiency wage theory. *Some* theory embodying payment of more than market-clearing wages is necessary to explain important features of the labor market, such as the existence of involuntary unemployment; the observed unwillingness of firms to reduce wages in the presence of job queues; the persistence of pay differentials for workers with seemingly identical characteristics; and the procyclical behavior of quits.

Alan Krueger and Lawrence Summers (1986a, b) and William Dickens and Lawrence Katz (1986a, b) have compiled impressive evidence in support of a central prediction of efficiency wage theory—that workers of identical characteristics receive different wages. These papers show the existence of large differences in wages across industries and occupations for workers after controlling for union status and observed worker and job characteristics. The impact of industry affiliation on wages is not just significant; it is also large, ranging in 1984 from a high of 38 percent above the mean for the petroleum industry to a low of 37 percent below the mean for private household services (Krueger-Summers, 1986a, p. 8). The leading neoclassical justifications for such differentials—unmeasured labor quality, compensating differentials, and transitory labor immobility—cannot easily account for the patterns of observed wage differentials.

As Lawrence Summers has observed, in its most general form, efficiency wage theory is virtually tautologous: it states that firms re-

frain from wage cutting because they don't perceive it to be in their interest. But *why* don't firms perceive wage cuts to be in their interest? The leading efficiency wage arguments are based on maximizing behavior; they stress the role of high wages both as a worker discipline device and also as a turnover reduction device. While efficiency wages per se are a useful concept, the maximizing models fail to explain four well-documented empirical regularities. They fail to predict wage compression (Robert Frank, 1984; Jean Grossman, 1983), the strong positive correlation between industry profits and industry wages (Dickens and Katz, 1986a; Krueger and Summers, 1986b), and the inverse correlation between unemployment and skill. Particularly striking is the unpredicted correlation of industry wage premia across occupations (Dickens-Katz, 1986b). *All* workers in better-paid industries tend to receive positive wage premia. That is, the wages of secretaries and engineers are highly correlated across industries. Ease of supervision and the magnitude of turnover costs might well be correlated across industries for a given occupation explaining, for example, why, say, skilled machinery operators receive positive wage premia in most industries. But there is no obvious reason why, say, secretaries, should be harder to supervise in the chemical industry where pay is high, than in the apparel industry where pay is low.

This paper presents an efficiency wage model based on fairness which provides natural explanations for the four regularities described above. According to this model, in industries where it is advantageous to pay some employees highly, it is considered *fair* to also pay other employees well. This theory is more in concert with basic worker motivation than standard maximizing models in which workers hedonistically maximize utility dependent only on their pay and their effort. Such hedonism seems too simplistic.

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Well-paid workers often grumble and shirk while poorly paid workers are often satisfied and hard working. All textbooks on compensation consider it self-evident that the most important aspect of a compensation system is its accordance with workers' conceptions of equity. Workers who consider themselves fairly treated are likely to work hard, and workers who consider themselves unfairly treated are likely to shirk.

Daniel Kahneman, Jack Knetsch and Richard Thaler (1986) have demonstrated that popular conceptions of fair wages and prices often diverge substantially from the prices of goods and the wages of labor which would clear competitive product and labor markets. A hardware store that raises the price of snow shovels after a snowstorm would be considered to have acted unfairly. Similarly, in the labor market, a firm that has been making money and that reduces the wages of its existing labor when there is unemployment in the area will also be considered to have acted unfairly.

Kenneth Arrow has written that economic theory is concerned with "the forces of greed and aggressiveness...not the best, but the strongest motives of humanity" (1972, p. 90). But, in contrast to the marketplace, where traders have little personal contact, in the workplace, where personal contact is close, other emotions such as "concern for fairness," pejoratively called "jealousy," are also important. The poets serve as witness to the importance of both greed (or desire) and jealousy (or hate). Following Robert Frost:

Some say the world will end in fire,
Some say in ice.
From what I've tasted of desire
I hold with those who favor fire.
But if I had to perish twice
I think I know enough of hate
To say that for destruction ice
Is also great
And would suffice.

I. Fairness and Unemployment

In the most rudimentary model of efficiency wages due to Robert Solow (1979), output with one unit of capital is a function

of labor efficiency units, which are the product of man-hours and effort expended, so that $q = f(e(w)l)$ where q is output; e is effort; w is the real wage; and l is man-hours. The optimal wage for the firm, that minimizes labor cost per efficiency unit, occurs at the wage w^* , where the effort function has unit elasticity with respect to the wage. With the wage at w^* , the firm will demand labor, l^* , up to the point where the marginal revenue product of labor equals the wage, w^* . In mathematics, this condition is $e(w^*)f'(e(w^*)l^*) = w^*$. If there are K units of capital and N units of labor inelastically supplied, unemployment will be $N - Kl^*$. The worker-discipline device and turnover models serve to derive the nature of the $e(w^*)$ function from first principles rather than to posit it in an *ad hoc* way.

An alternative efficiency wage model that considers fairness could be formulated as follows. Suppose that there are two types of labor, call them l_1 and l_2 , and suppose also that output with one unit of capital can be expressed as: $q = e(\cdot)f(l_1, l_2)$. Let us follow Solow in trying on for size an *ad hoc* assumption about the effort function $e(\cdot)$, whose argument is as yet unspecified. Assume that e depends on the variance of wages paid by the representative firm: $e = e(\sigma^2(w))$. The simple rationale for this dependence of effort on wage variance is that firms with less variance in their compensation will have more harmonious labor relations and thus achieve higher output per worker. The firm's profit function is now: $\pi = e(\sigma^2(w))f(l_1, l_2) - w_1l_1 - w_2l_2$.

It is easy to see that in such an economy the higher paid labor will always receive a market-clearing wage and will have no involuntary unemployment. A reduction in the wage for higher-paid employees unambiguously benefits the firm. As the wage to higher-paid workers declines, effort increases as a consequence of the decrease in the variance of wages; in addition, the firm's wage bill declines. Without loss of generality, we may assume that l_1 is the higher-paid labor type and that its wage clears the labor market. The firm will then choose l_1 and w_1 at market-clearing levels, and it will choose l_2 and w_2 to maximize profits. With

an interior solution to both of these maximization problems, it will choose: $\partial\pi/\partial l_2 = 0$ and $\partial\pi/\partial w_2 = 0$. If these two conditions have an interior solution and $N_2 > Kl_2^*$, there will be unemployment of type 2 labor. The wage differential between types 1 and 2 labor will be "compressed" in comparison with the perfectly competitive equilibrium.

A very slight change in the model yields a result exactly analogous to Solow's. If one assumes that the effort of the lower-paid labor depends on the wage gap, but the effort of the higher-paid labor is independent of the gap, the firm then maximizes $\pi = f(l_1, e(\sigma^2(w))l_2) - w_1l_1 - w_2l_2$. Here w_1 and l_1 are at market-clearing levels. And the firm's optimal choice of effort and wage for type 2 labor satisfies the familiar Solow condition. With an interior-maximizing solution, the elasticity of effort with respect to the wage of type 2 labor (due to the dependence of effort on the variance of wages) will be unity.¹

This model provides a straightforward rationale for wage compression and easily explains the observed positive correlation of industry wage premia across occupations. In industries where market forces compel firms to pay high wages to their engineers, considerations of equity will generate high pay to secretaries as well. The model also offers an explanation why unskilled workers have much higher unemployment rates than skilled workers. The model predicts that groups of workers with high pay will have lower unemployment than groups with low pay. Because high skill and high pay are typically correlated, this means that higher-skilled groups will usually have lower unemployment rates than lower-skilled groups. Occasionally, however, due to oversupply, highly skilled groups have high unemployment rates. Classical musicians are an example. Despite their skill, classical musicians are in considerable supply, and, as a result, their pay is low. They also have high rates of unemployment. Our model, which pre-

dicts that low-paid workers will experience high unemployment, is consistent with this anomaly.

The preceding model focuses on the adverse reaction of workers to inequitable pay disparities among employees. The same destructive forces of jealousy will also operate when there are "unfair" disparities between the earnings of the firm and the earnings of workers. Thus, the natural extension of our model, to include a relation between effort, firms' earnings, and workers' earnings, predicts the observed positive correlation between profits and wages. The correlation between profits and wages gives added reason for the positive correlation of industry wage premia across occupations.

II. The Fair Wage/Effort Hypothesis

The Solow model is useful in characterizing the type of unemployment which occurs when the labor market is in equilibrium. Nevertheless, it is not a complete model because the relation $e = e(w)$ is *ad hoc*. Similarly, the effort/wage-variance model predicts an unemployment equilibrium which accords with the stylized facts of the labor market discussed above. But this model, like the Solow model, lacks an adequate micro-foundation for its effort function.

Social exchange theory in sociology and equity theory in psychology both provide a rationale for a precise relationship between effort and fairness. (For a similar rationale based on worker cohesion, see David Levine, 1987.) According to equity theory, in interpersonal exchange, the ratio of the perceived value of outcomes to inputs will tend to be equal. It is important to note that this equality is based on *subjective valuations* and not on *market valuations*.

In the context of a wage contract, equity theory suggests that the perceived value of an individual's labor input, $e\hat{w}$, should equal the perceived value of the compensation received, w , where w is the actual wage and $\hat{w}$ is the wage which is considered fair compensation for "normal" effort. Assuming that workers supply "normal effort," $e = 1$, if they are compensated at least fairly, and lower effort if they receive "unfair wages," equity

¹This condition assumes that $\sigma^2(w)$ is calculated with constant amounts of l_1 and l_2 labor. If $\sigma^2(w)$ is calculated on the assumption of varying l_1 and l_2 , no simple formula obtains.

theory implies that $e = \min(w/\hat{w}, 1)$. Insofar as the actual wage w falls short of the fair wage $\hat{w}$, the worker will withdraw effort so as to maintain equity between the subjective value of input and the subjective return. We call the preceding behavior with respect to effort the *fair wage/effort hypothesis*. Under this hypothesis, if production depends on labor efficiency units and if firms have even the smallest preference for paying the fair wage (rather than less than fair wages), firms will always find it worthwhile to pay at least the fair wage $\hat{w}$.

There are many possible theories concerning the determination of $\hat{w}$; but any theory which leads to an equilibrium value of $\hat{w}$ for some class of labor greater than the market-clearing level will result in unemployment. The work previously mentioned of Kahneman et al. tells us that such discrepancies do occur; popular views of fair wages do not coincide with market clearing. A theory, for example, in which workers use the pay of higher-paid workers at the same firm as their anchor in determining fair pay, gives exactly the type of result obtained in the last section. Lower-paid workers are unemployed while higher-paid workers receive market-clearing wages.

The fair wage/effort hypothesis serves three useful purposes. First, it gives a natural functional form for the relation between effort and wages—while leaving considerable room for variations in the theory dependent on the determination of the fair wage. Second, it grounds the wage-effort function in theory from two social sciences, both reflecting the basic human desire to “*get even*.” Third, equity theory (and to a lesser extent social exchange theory) has been subject to empirical verification: for the psychologists through experiments, and for the sociologists through field studies.

A number of experiments have been performed to test the implications of equity theory for the inequity-performance relationship. Surveys of these experiments suggest that they are usually supportive of equity theory. In a typical study, subjects are asked to perform a task, such as proofreading, and offered an “hourly” or “piece” rate designed to induce a clear feeling of over- or under-

payment relative to a salient comparison group. The outcomes of these experiments typically accord with the predictions of equity theory: underpaid hourly subjects usually decrease their labor input to achieve an input-outcome balance. For example, Edward Lawler and Paul O’Gara (1967) compared the performance of workers who were paid the “going piece rate” vs. an underpaid rate for conducting interviews. They found the lower-paid workers produced work of substantially lower quality (but greater quantity) and had reduced self-esteem. Further evidence comes from a study by Robert Pritchard, Marvin Dunnette, and Dale O. Jorgenson (1972), who hired men to work for a manpower firm they set up. After the workers had been on the job for three days, the firm changed the pay with some subjects receiving increases and some subjects receiving decreases in pay. Those who got pay cuts were angry and performed less well in their work. All of these experiments suffer from two unavoidable difficulties: the anchor point of the fair wage must be inferred and the experiment by its nature is of short duration. Whether the observed behavior consistent with the fair wage/effort hypothesis will continue over long periods of time has not been inferred from experimental data.

Sociologists have explained field study observations with social exchange theory. The most famous of these is a study by Peter Blau (1955). In a federal bureaucracy agents varied in expertise. He found that agents of equal expertise frequently consulted one another, but only rarely did agents of average expertise consult the experts. Blau asked why the agents of average expertise did not consult the experts more frequently. The experts were observed to be friendly and always helpful, if asked. According to Blau’s answer, the two sides of the exchange had equal value—the average consulting agent receiving the advice of the expert with the expert receiving gratitude and respect. But such gratitude and respect had diminishing returns. Equilibrium occurred where the exchange was considered fair by both sides; the payment in gratitude and respect to the experts was of equal value with the aid given by the experts.

The scientific findings of social science notwithstanding, the key advantage of equity theory is its accordance with common sense. According to George Homans: "The more to a man's disadvantage the rule of distributive justice fails of realization, the more likely he is to display the emotional behavior we call anger" (1961, p. 75). Or, more prosaically, if people do not get what they think they deserve they get mad. This is the basic simple proposition underlying the fair wage/effort hypothesis. If w falls short of $\hat{w}$, people will be angry. As a result of this anger, their effective labor input is reduced below the level they would put in if fully satisfied.

III. Theories of Fairness

The fair wage/effort hypothesis specifies the relation between effort and wages conditional on the fair wage $\hat{w}$. Different theories of the fair wage $\hat{w}$, will yield different theories of unemployment. Our longer paper (1987) constructs a model with two groups of workers, a high-paid group and a low-paid group. The low-paid group regard their fair wage as an average of the wages of the high-paid group in the same firm (who constitute part of their reference group) and the wages they would be paid if the market cleared. This formulation is a compromise between theories in which market forces completely determine fairness (in which case fair wages and market-clearing wages would exactly coincide) and theories in which sociological theories completely determine fairness (in which case the wages of some reference group would determine the fair wage). The micro model is surprisingly complex, but the macro intuition of the variance-efficiency wage theory of the last section survives. There is wage compression and the low-paid workers are those with unemployment.

The assumption that low-paid workers base their notions of fairness on the pay received by more highly skilled workers is undoubtedly a substantial simplification of how individuals realistically form conceptions of equity. Most workers in fact feel that fairness requires a relation between remuneration and performance. Workers with

low skill do not consider it fair to receive the identical wage as workers who are obviously more skilled. The theory of unemployment described here does not, however, rely on the simplistic assumption that workers view wage equality as fair. What is important is that workers regard a fair wage system as one with pay differentials which are more compressed than productivity differentials. An interesting study by Herbert Meyer (1975) suggests that this is indeed the case. He found in surveys that workers have a systematic tendency to view their own performance as superior. In four surveys he found that between 68 and 86 percent of workers viewed their performance in the top quartile.

IV. Conclusion

Careful use of sociological theory, which thankfully also coincides with common sense, produces efficiency wage models of unemployment. The key to such models is the fair wage/effort hypothesis, that says that effort is proportional to the wage for workers paid less than the subjectively determined fair wage. The fair wage/effort hypothesis comes naturally from equity theory and social exchange theory. A simple hypothesis regarding the determination of the fair wage $\hat{w}$ explains why unemployment is greater for low-paid labor groups (even of high objective skill). Efficiency wage theories based on fairness do not suffer from the criticism levelled at more neoclassical theories that if all agents are fully maximizing more complicated contracts will eliminate involuntary unemployment. They are also consistent with observed industry and occupational wage differentials.

REFERENCES

- Akerlof, George and Yellen, Janet, "The Fair Wage/Effort Hypothesis and Unemployment," mimeo., University of California, Berkeley, July 1987.
- Arrow, Kenneth, "Models of Job Discrimination," in A. H. Pascal, ed., *Racial Discrimination in Economic Life*, Lexington: D. C. Heath, 1972, 83-102.
- Blau, Peter M., *The Dynamics of Bureaucracy*:

- A Study of Interpersonal Relations in Two Government Agencies*, Chicago: University of Chicago Press, 1955.
- Dickens, William and Katz, Lawrence, (1986a) "Interindustry Wage Differences and Industry Characteristics," NBER Working Paper No. 2014, September 1986.
- _____ and _____, (1986b) "Industry and Occupational Wage Patterns and Theories of Wage Determination," mimeo., March 1986.
- Frank, Robert, "Are Workers Paid their Marginal Products?," *American Economic Review*, September 1984, 74, 549-71.
- Grossman, Jean, "The Impact of the Minimum Wage on Other Wages," *Journal of Human Resources*, Summer 1983, 18, 359-78.
- Homans, George C., *Social Behavior: Its Elementary Forms*, New York: Harcourt Brace Jovanovich, 1961.
- Kahneman, Daniel, Knetsch, Jack and Thaler, Richard, "Fairness as a Constraint on Profit Seeking: Entitlements in the Market," *American Economic Review*, September 1986, 76, 728-41.
- Krueger, Alan and Summers, Lawrence, (1986a) "Efficiency Wages and the Inter-Industry Wage Structure," mimeo., 1986.
- _____ and _____, (1986b) "Reflections on the Inter-Industry Wage Structure," Harvard Institute of Economic Research Discussion Paper No. 1522, July 1986.
- Lawler, Edward E. and O'Gara, Paul W., "The Effects of Inequity Produced by Underpayment on Work Output, Work Quality, and Attitudes Toward the Work," *Journal of Applied Psychology*, October 1967, 51, 403-10.
- Levine, David, "Cohesiveness and the Inefficiency of the Market Solution," mimeo., Harvard University, March 1987.
- Meyer, Herbert, "The Pay for Performance Dilemma," *Organizational Dynamics*, Winter 1975, 3, 39-50.
- Pritchard, Robert D., Dunnetter, Marvin D. and Jorgenson, Dale O., "Effects of Perceptions of Equity and Inequity on Worker Performance and Satisfaction," *Journal of Applied Psychology Monograph*, February 1972, 56, 75-94.
- Solow, Robert, "Another Possible Source of Wage Stickiness," *Journal of Macroeconomics*, Winter 1979, 1, 79-82.